



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2021-22
MONDAY TEST

CLASS: IX
SUBJECT: PHYSICS

FULL MARKS: 40
DATE: 13.07.2021

General Instructions:

- This paper consists of two printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.
- Marks will be deducted for spelling errors.

Question 1

[2+2+2+2+2]

- (a) Can we have an eastward velocity but a westward acceleration? Give your comments for your answer.
- (b) Express the derived units of (i) watt and (ii) pascal in terms of kilogram, metre and second.
- (c) What is the ratio of the acceleration due to gravity at two places where the ratio of time period of the pendulum is 2: 3?
- (d) On what principle does a screw gauge work? Express the least count of a micrometer screw gauge in cm.
- (e) What are the functions of vernier scale and strip in a vernier calipers?

Question 2

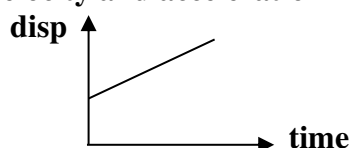
[2+2+2+2+2]

- (a) Which quantity does the speedometer of a vehicle measure? Define this quantity.
- (b) Why do two bodies of different masses released from same height in vacuum reach the ground simultaneously?
- (c) Draw the velocity – time graph of a freely falling body. How can you find the acceleration of the body from the graph?
- (d) Write any two properties of a unit to measure a physical quantity.
- (e) Which unit is used to find the mass of large heavenly bodies? How is this related to the SI unit of mass?

Question 3

[3 +4 +3]

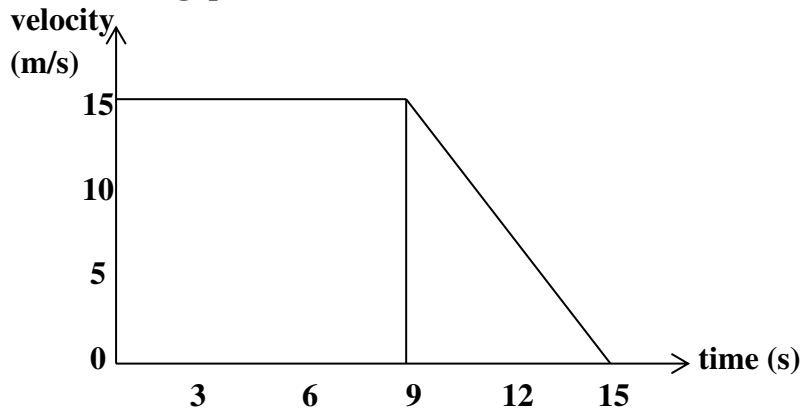
- (a) A car is moving in a straight line with a speed 36 km/h initially. Brakes are now applied and the car stops in 5 s. Find the speed in m/s and retardation produced. What will be the speed of the car after 3 s of applying the brakes?
- (b) Why do we measure the time period for more than one oscillation? Which type of pendulum is used in a clock? What is its feature?
- (c) Can we find the velocity and acceleration from the following graph? Explain.



Question 4

[4+3+3]

- (a) The motion of a car is as shown. Study the velocity-time graph and answer the following questions



- (i) Find out the acceleration for the first 9 s.
 - (ii) Determine the retardation in the last 6 s.
 - (iii) What is the total distance travelled?
- (b) Prove the relation $s = ut + \frac{1}{2}at^2$ either graphically or by analytical method.
- (c) A body moving with constant acceleration travels 3m and 8m in 1 s and 2s respectively. Find the initial velocity and the acceleration.